

Andrea Galenda

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EDUCATION

MSc. in Mechanical Engineering, Mechatronics and Robotics | Politecnico di Milano Sept. 2023 – Present
Coursework: Mechanical Systems Dynamics, Signal Processing, Actuating Devices | **GPA:** 29.2/30

Erasmus+ Program, Electronics and Robotics Engineering | Polytech Paris-Saclay Sept. 2024 – Feb. 2025

BSc. in Mechanical Engineering | Politecnico di Milano Sept. 2020 – Sept. 2023

EXPERIENCE

NASA Jet Propulsion Laboratory Dec. 2025 – Present
Visiting Student Researcher - Aerial Robotic Systems (Section 358M) Pasadena (CA), United States

- Multi-physics modeling: developed a 6-DOF flight dynamics, mass transfer and thermal model for a Venus Aerobot
- Disturbance modeling and analysis: modeled critical structural, atmospheric and environmental perturbations, including gas leaks, thermal flux variations and flow disturbances
- Simulated 3D Aerobot trajectories and thermal behavior (Python/MATLAB), evaluating the effects of the disturbances on the mission performance

PROJECTS

Mechanism Design and Robotic Manipulation

- Designed and 3D-printed a custom support structure and a motor-driven rack-and-pinion end-effector to autonomously manipulate complex mechanical closures
- Developed an automated path-planning sequence in Python, optimizing point-to-point and spline motions to execute precision manipulation tasks
- Integrated a real-time fault detection system utilizing robot-mounted camera vision (TMFlow color recognition) and torque feedback to verify task completion and ensure safe operation

Bayesian Optimization for Robot Controller Tuning

- Formulated a two-phase Bayesian Optimization (BO) algorithm to autonomously estimate unknown link masses and tune PID controller gains for a 7-DOF robotic manipulator
- Designed a multi-objective cost function to minimize position/velocity tracking errors while enforcing strict constraints on torque saturation and movement smoothness

Reduced Order Modeling for Structural Elasticity

- Developed Data-Driven Reduced Order Models (ROM) using Singular Value Decomposition (SVD) and POD-Galerkin to simulate structural elasticity and deformation under dynamic loads
- Designed and trained deep neural networks (POD-NN and DL-ROM autoencoders) to map high-dimensional physics parameters to a latent space, reducing simulation compute times from seconds to milliseconds

Redesign of a Scissor Lift System (BSc Thesis)

- Conducted the mechanical redesign of a scissor lift system for radiography tables, successfully reducing manufacturing and assembly costs by 20%
- Modeled and verified critical structural components through Finite Element Analysis (FEA) to ensure safety under 300 kg static and 200 kg dynamic loads
- Optimized the actuation mechanism by integrating a horizontal linear actuator with parallel compression springs to minimize peak torque requirements

SKILLS

Simulation and Physics: MATLAB/Simulink, ROS, Gazebo, Abaqus (FEA)
Data Analysis and Scripting: Python, C++, Signal Processing Toolbox (MATLAB)
CAD: SolidWorks, CATIA, Autodesk Inventor, Onshape, Solid Edge
Hardware: 3D Printing, Arduino, PCB Assembly
Languages: English (C1), French (B2 - Reading-Listening-Speaking), Italian (Native)